Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18ME56]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

METROLOGY AND MEASUREMENTS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. What is metrology? State and explain the objectives of metrology. **07**

b. Discuss the need of having light wave as standard of measurement. **07**

2. a. Sketch and explain the working of sigma comparator

b. Sketch and explain sine bar highlighting its applications

3. a. Explain with an example the various stages of generalized measurement system. **07**

b. Explain the terms **07**

- **i.** Accuracy
- **ii.** Precision
- **iii.** Repeatability
- **iv.** Resolution

4. a. Explain the following concepts: **08**

- i. Hysteresis
- ii. Loading effect
- iii. Linearity in measurements
- iv. Delay

b. Explain errors in measurements. Classify errors. **06**

5. a. With the help of neat sketch explain the working principle of prony brake dynamometer. **06**

b. Describe with a neat sketch McLeod gauge. **08**

OR

6. a. With a neat sketch, explain pressure measurement using pirani gauge. **08**

b. Explain briefly, the working principle of platform balance. **06**

7. a. State the laws governing the functioning of thermocouples. **07**

b. Describe the construction and working of radiation pyrometer. **07**

OR

8. a. Explain with a sketch bonded type resistance strain gauge. **06**

b. Derive the expression for the gauge factor of the strain gauges in terms of Poisson's ratio. **08**